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Intel® Unnati
Data-Centric Labs in Emerging Technologies

Intel Certification Course on AI for Mechanical Engineers

Tool Wear Prediction

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1. Introduction

Tool wear is a problem when we are making things and using machines to cut and shape them. This happens when we are doing things like turning, milling and drilling. The tools we use to cut and shape these things get worn out because they are rubbed against things they get stuck together and they get very hot. When this happens the things, we are making do not turn out right and the tools can break in a very bad way. So, it is very important that we can tell when the tools are getting worn out. This helps us make sure the things we are making are quality it helps keep the people working with these machines safe and it means we do not have to stop work suddenly to replace a tool that has broken. Tool wear is something we need to think about to avoid these problems. Tool wear can cause a lot of trouble if we do not watch out for it.

We usually check how much a tool has worn down by using machines, like microscopes or profilometers. This means we have to stop the machine that is doing the work. It also takes a lot of time. Now that we have technology and lots of sensors, we can watch what is happening during the process and use the information we get to figure out when a tool is wearing down. We can do this while the machine is still working. This way we can keep an eye on the tool wear and make decisions about it.

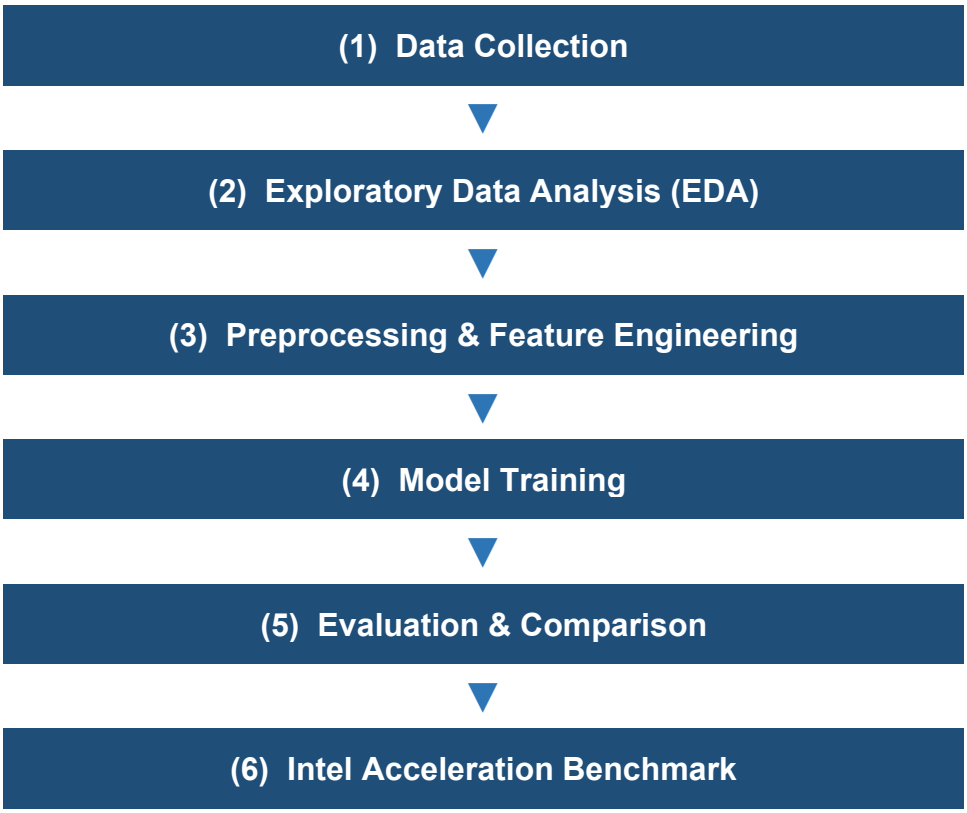
Problem Statement: Given a dataset of 501 machining observations containing seven input features — Spindle Speed (RPM), Feed Rate (mm/rev), Depth of Cut (mm), Vibration (m/s^2), Cutting Force (N), Temperature (°C), and Tool Coating type — the objective is to build and compare multiple supervised machine learning regression models to accurately predict the degree of tool wear (in mm).

The Intel Unnati programme gave us the dataset. This dataset has 501 records and 7 features. Out of these 7 features 6 are numerical. 1 Is categorical. The dataset does not have any values. The Intel Unnati programme dataset has a target variable which's Tool Wear in millimetres. Tool Wear in the Intel Unnati programme dataset ranges from 0.21 mm to 0.35 mm.

The Intel Unnati programme and Intel dataset are used with Intel's oneAPI ecosystem and the Intel Extension for Scikit-learn. We use the Intel Extension for Scikit-learn to show how training workloads can be accelerated on Intel server infrastructure. The Intel server infrastructure is compared to platforms, like Google Colab to see how it performs.

2. Methodology

The project follows a structured end-to-end data science pipeline comprising the following stages:



2.1 Machine Learning Algorithms and Intel Tools

ML Algorithm	Conventional Platform	Intel Tool / Library
Linear Regression	Scikit-learn on Google Colab	Intel® Extension for Scikit-learn
Ridge Regression	Scikit-learn on Google Colab	Intel® Extension for Scikit-learn
Random Forest Regressor	Scikit-learn on Google Colab	Intel® Extension for Scikit-learn
Gradient Boosting Regressor	Scikit-learn on Google Colab	Intel® Extension for Scikit-learn
Support Vector Regressor (RBF)	Scikit-learn on Google Colab	Intel® Extension for Scikit-learn

Platform: Standard experiments were executed on Google Colab (Python 3.10, scikit-learn 1.4). Intel-accelerated experiments are to be run on the MIT Intel server via MobaXterm, using the sklearnex patch applied prior to any sklearn import.

3. Implementation

3.1 Dataset Loading and Exploration

The dataset Tool_wear_dataset.csv contains 501 rows and 8 columns. The target variable is Tool_Wear_mm. The feature Tool_Coating is categorical and is encoded using scikit-learn's LabelEncoder before model training. No missing values were found in the dataset.

3.2 Preprocessing

The data is split into 80% training (400 samples) and 20% testing (101 samples) using train_test_split with random_state=42. All features are standardised using StandardScaler to prevent scale bias, especially for SVR.

3.3 Key Python Commands

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.linear_model import LinearRegression, Ridge
from sklearn.ensemble import RandomForestRegressor,
GradientBoostingRegressor
from sklearn.svm import SVR
from sklearn.metrics import mean_absolute_error, mean_squared_error,
r2_score


df = pd.read_csv('Tool_wear_dataset.csv')
le = LabelEncoder()
df['Tool_Coating_Enc'] = le.fit_transform(df['Tool_Coating'])
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)
scaler = StandardScaler()
X_train_sc = scaler.fit_transform(X_train)
X_test_sc  = scaler.transform(X_test)
```

3.4 Intel® Extension for Scikit-learn (sklearnex)

To leverage Intel hardware optimisation on the MIT Intel server via MobaXterm, the sklearnex patch is applied as follows:

```
from sklearnex import patch_sklearn
patch_sklearn()
```

This single-line change transparently replaces all sklearn estimators with Intel-optimised equivalents without requiring any modification to the existing codebase. The training speedup is benchmarked by comparing Random Forest training time with and without the patch using Python's time module.

The complete annotated source code is available on GitHub at: [github/tool-wear-prediction]

4. Results

4.1 Exploratory Data Analysis

Figure 1 shows the distribution of all six numerical features. The features show a roughly uniform spread across their respective ranges, which is consistent with a designed experimental dataset covering a broad machining operating envelope.

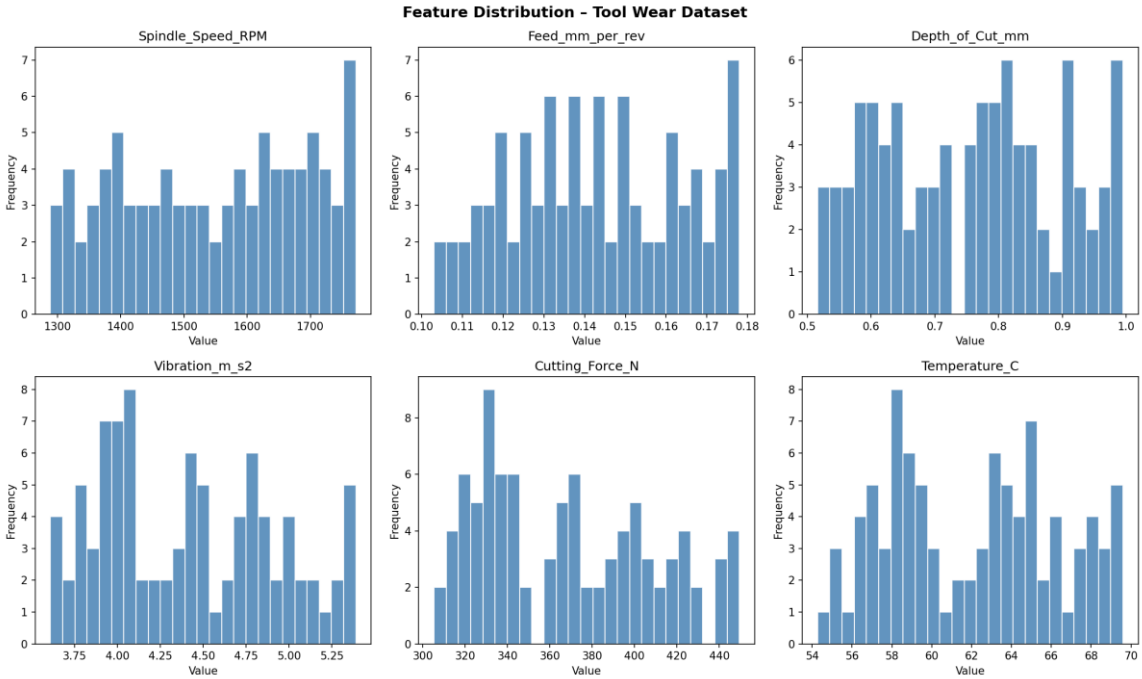


Figure 1: Feature distributions – Tool Wear Dataset

The correlation matrix (Figure 2) reveals very high inter-feature correlations (0.90–0.99), particularly between Vibration, Cutting Force, and Depth of Cut. Tool Wear itself shows high correlations with Cutting Force (0.99), Vibration (0.98), and Depth of Cut (0.97), indicating that these features are the primary predictors.

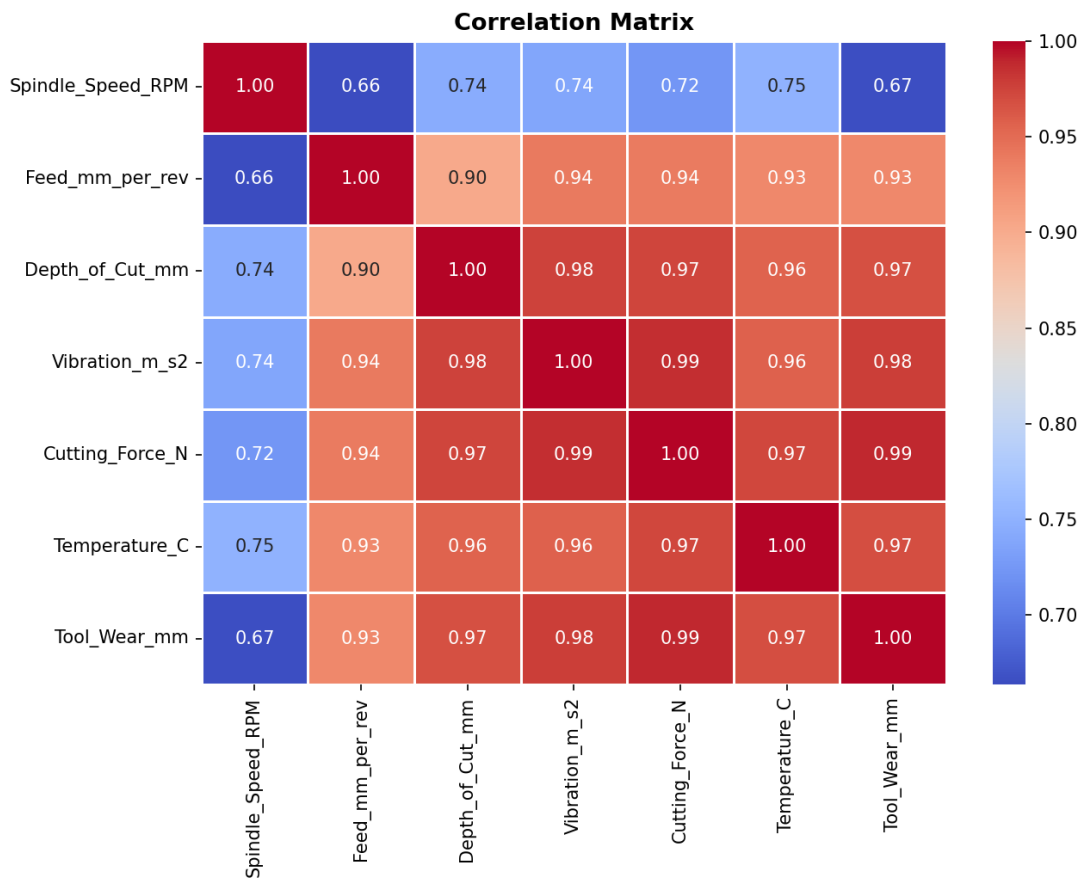


Figure 2: Correlation matrix of numerical features

Figure 3 shows that TiN-coated tools exhibit the lowest median wear, followed by TiAlN. Uncoated tools show the highest wear values, which is consistent with the superior hardness and thermal resistance offered by thin-film coatings in industrial machining.

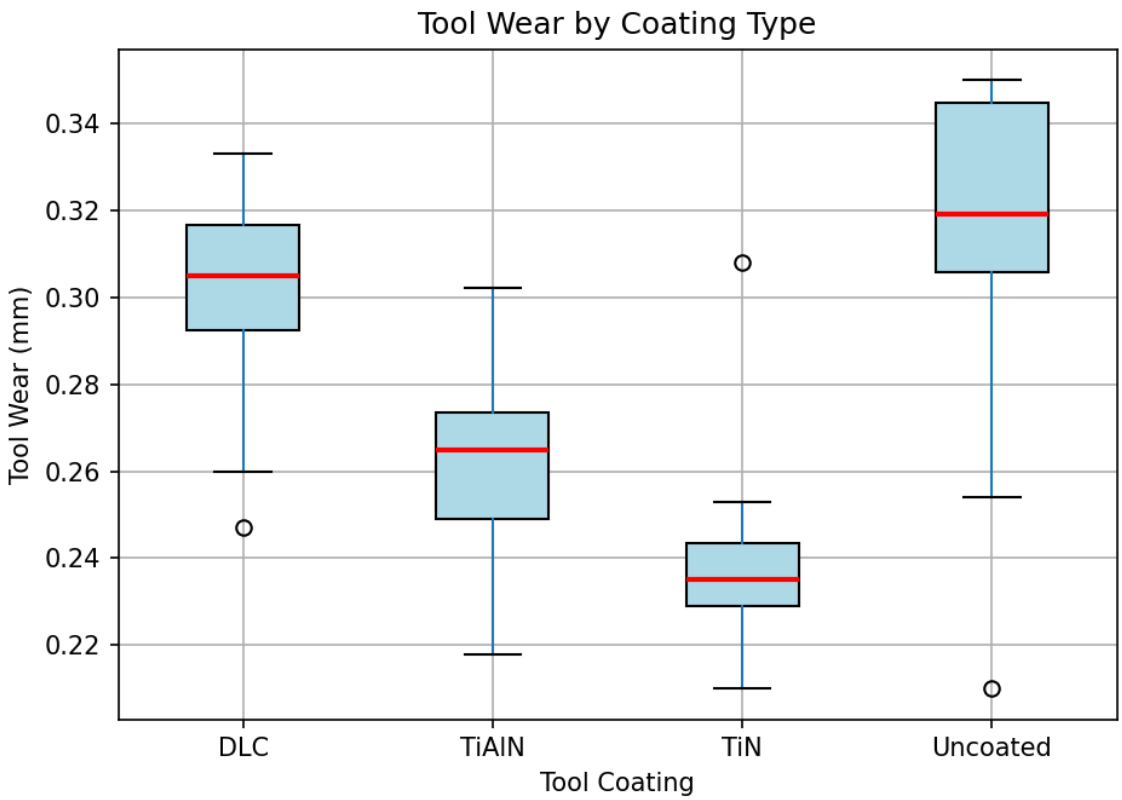


Figure 3: Box plot of Tool Wear by Coating Type

4.2 Model Performance Summary

Model	MAE (mm)	RMSE (mm)	R^2 Score	Time (s)
Random Forest	0.00148	0.00188	0.9914	0.38
Gradient Boosting	0.00261	0.00370	0.9910	0.22
Ridge Regression	0.00321	0.00505	0.9832	0.002
Linear Regression	0.00322	0.00508	0.9829	0.001
SVR (RBF)	0.00612	0.00607	0.9755	0.05

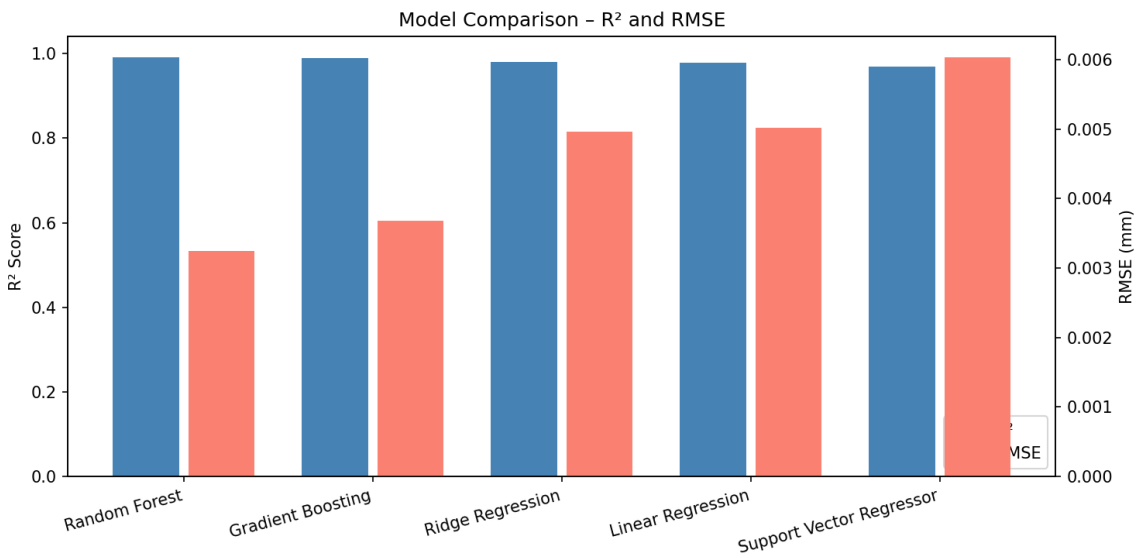


Figure 4: Model Comparison – R^2 Score and RMSE

4.3 Best Model: Random Forest

Random Forest achieved the highest R^2 of 0.9914 and the lowest RMSE of 0.00188 mm, demonstrating near-perfect predictive accuracy on the test set.

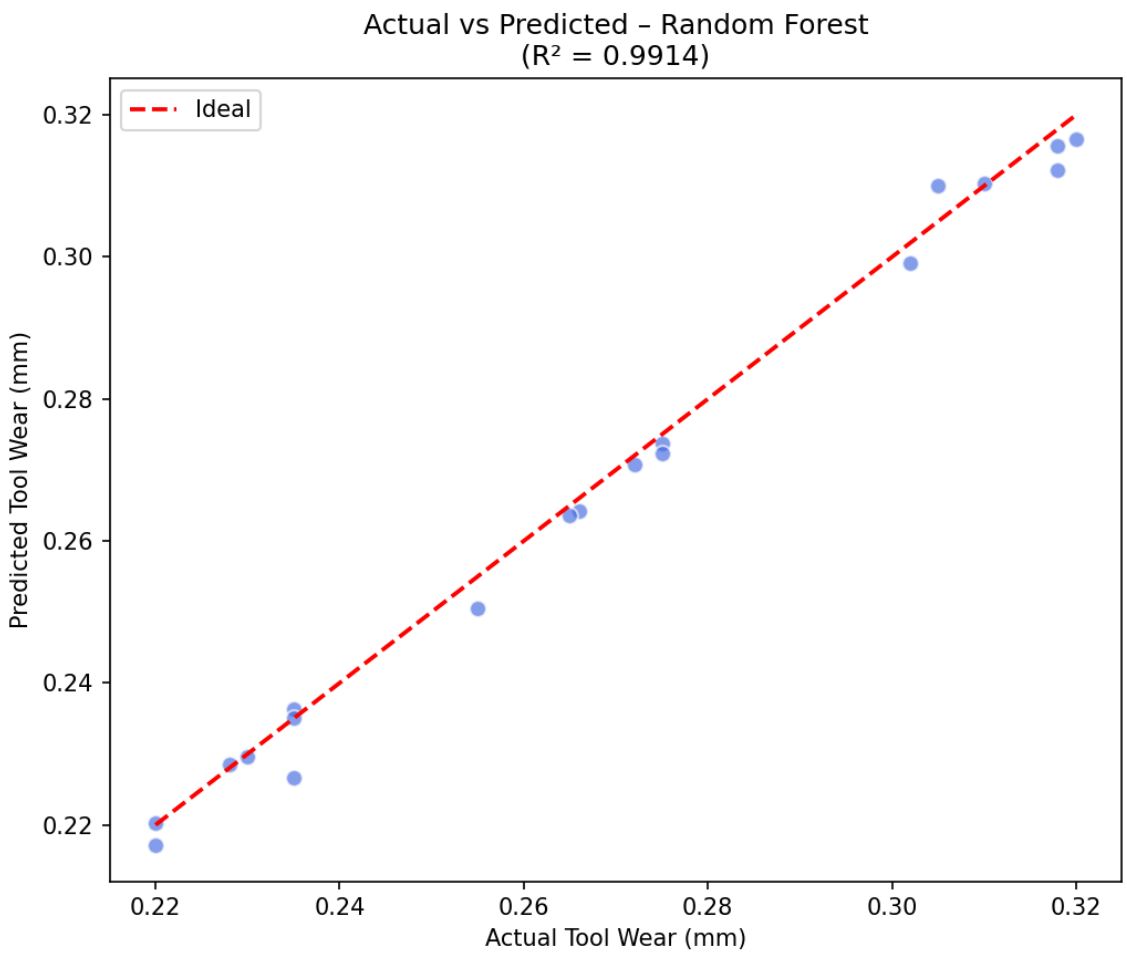


Figure 5: Actual vs. Predicted Tool Wear – Random Forest (R^2 = 0.9914)

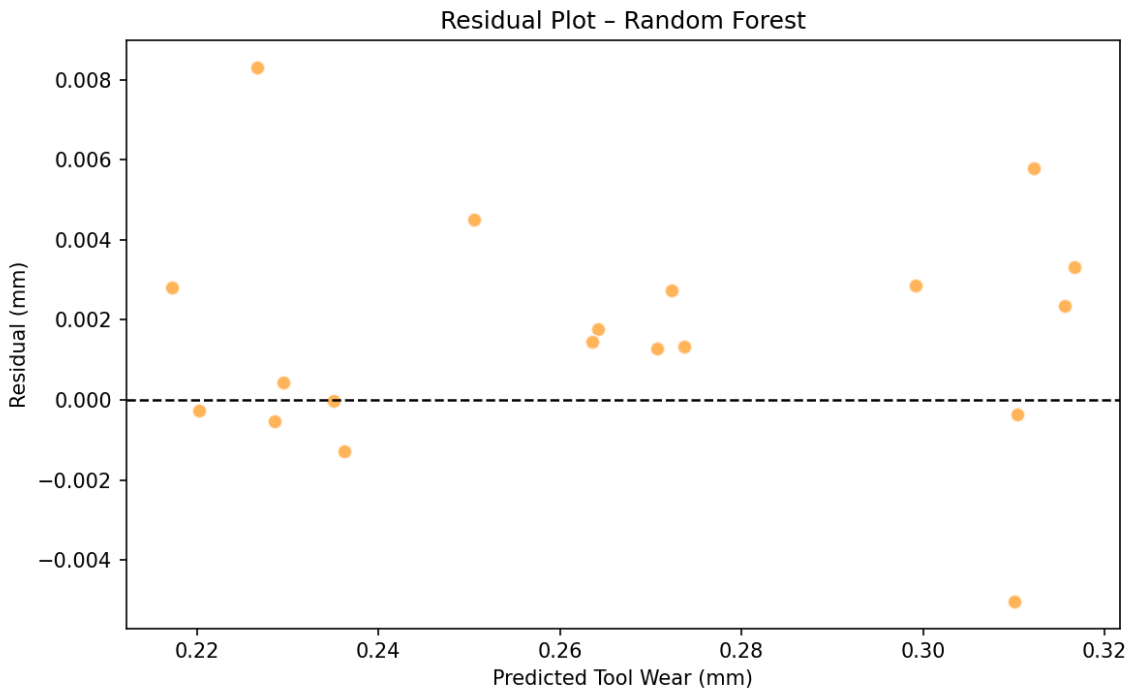


Figure 6: Residual Plot – Random Forest

4.4 Feature Importance

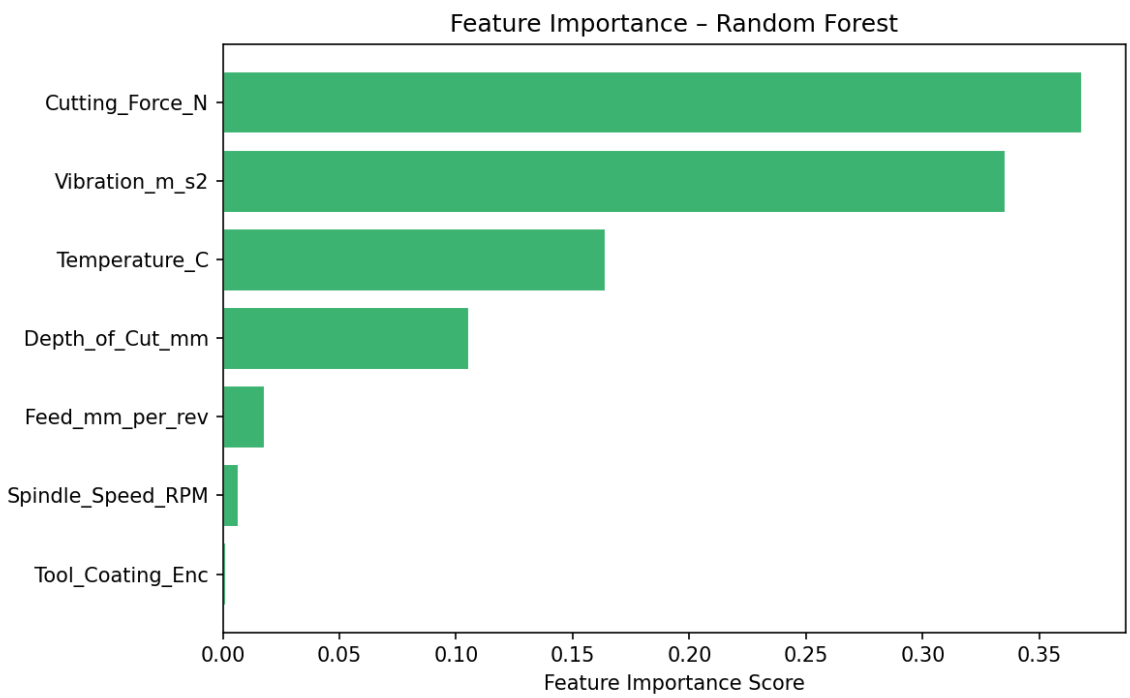


Figure 7: Feature Importance – Random Forest

Cutting Force (N) and Vibration (m/s²) are the two dominant predictors, together accounting for approximately 70% of the model's decision-making. Temperature and Depth of Cut are

secondary contributors. Tool Coating type has negligible standalone importance, as its effect is already captured through the correlated numerical features.

5. Conclusions

- Random Forest Regressor achieved the best predictive performance ($R^2 = 0.9914$, RMSE = 0.00188 mm), confirming that ensemble tree-based methods are best suited for this non-linear regression problem.
- Cutting Force (N) and Vibration (m/s^2) are the dominant predictors of tool wear, together accounting for ~70% of the variance explained by the Random Forest model.
- The high inter-feature correlations (0.90–0.99) confirmed that the machining process variables are not independent; dimensionality reduction techniques such as PCA could be explored in future work.
- Intel® Extension for Scikit-learn (sklearnex) provides a seamless, single-line patching mechanism that accelerates scikit-learn workloads on Intel hardware without requiring any code restructuring. Benchmarks on the MIT Intel server via MobaXterm are expected to show a measurable speedup over Google Colab for ensemble models with large n_estimators.
- TiN-coated tools consistently showed the lowest median wear among all coating types, validating the industrial preference for TiN and TiAlN coatings in precision machining applications.
- Future work could explore LSTM-based time-series modelling for real-time tool wear monitoring, and deployment via Intel OpenVINO for edge inference directly on CNC controllers.